

Align Only Where Needed: Alignment-Free Phoneme Scoring and Embedded CTC–Viterbi Stress Assessment

Long, Nguyen Trung
long.nt270504@gmail.com

Abstract—Phoneme scoring does not necessarily require phone boundaries, whereas syllable-stress features do require temporal localization. We therefore combine Cao et al.’s alignment-free Connectionist Temporal Classification (CTC) vector features and a phone-only GOPT model with an embedded, transcript-conditioned CTC–Viterbi path used only by a Mallela-inspired sequential stress scorer. We compare this proposal on the official speaker-disjoint SpeechOcean762 split with (i) a modular Montreal Forced Aligner (MFA) pipeline and (ii) Gong et al.’s joint multi-task GOPT architecture, adapted to the same CTC front end. Across five independent seeds and 46,119 common test phones, the proposed phone branch obtains Pearson correlation 0.645 ± 0.009 , versus 0.413 ± 0.013 for the MFA modular baseline and 0.418 ± 0.005 for joint GOPT. On 2,354 common polysyllabic words, CTC–Viterbi and MFA yield stress-correctness AUROC 0.681 and 0.662. Their canonical stress-location accuracies are 85.14% and 85.85%; the paired difference is -0.70 percentage points (95% speaker-bootstrap interval $[-1.81, 0.48]$), so no stress difference is statistically resolved. Complete single-request CPU latency is 212.6 ms for the proposal, 234.6 ms for modular MFA, and 192.5 ms for joint GOPT. Thus the hybrid design is $1.10\times$ faster than the like-for-like modular MFA pipeline and substantially improves phone correlation, while the joint model exposes a different speed–accuracy trade-off. We release reproducible feature, training, paired-stress, latency, and official-checkpoint validation code.

Index Terms—pronunciation assessment, goodness of pronunciation, CTC, GOPT, lexical stress, forced alignment

I. Introduction

Computer-assisted pronunciation training benefits from segmental feedback about individual phones and suprasegmental feedback about lexical stress. Traditional goodness-of-pronunciation (GOP) pipelines first align a known transcript, then aggregate acoustic evidence within each phone interval [1], [2]. Stress systems likewise use syllable or vowel regions to measure duration, intensity, fundamental frequency, and spectral prominence [3], [4]. This often leads a complete assessment service to run an external forced aligner even when only one branch truly needs boundaries.

Cao et al. recently showed that CTC permits phone-level pronunciation features without selecting a frame path [5]. Their alignment-free vector marginalizes over every CTC path for the canonical sequence and locally modified alternatives. Conversely, the sequential stress models of Mallela et al. operate on syllable features, so they still need localized acoustic evidence [6]. Our design follows

both observations: do not align for the phone score, but derive stress regions from a Viterbi path already available in the resident CTC posterior matrix.

The complete proposal has no MFA/Kaldi request-time dependency. Its Viterbi trellis is nevertheless conditioned on the canonical phone sequence; we call it embedded localization, not unconstrained recognition. The relevant engineering distinction is that the same CTC inference supplies both Cao features and stress regions, without a second acoustic model, dictionary pipeline, database, or alignment subprocess.

We evaluate three complete systems (Fig. 1):

- 1) a strong modular MFA baseline with an MFA-aligned CTC vector phone scorer and an MFA-conditioned sequential stress scorer;
- 2) the proposed Cao alignment-free phone scorer plus the same stress scorer using embedded CTC–Viterbi intervals; and
- 3) Gong et al.’s published multi-aspect, multi-granularity GOPT architecture [7], adapted to emit both phone and word stress scores from the controlled MFA-aligned CTC input.

The study reports a common held-out subset, five initialization seeds, a paired speaker bootstrap for the fixed stress branches, and one-request latency with equal thread/job settings. We also run the unmodified official GOPT checkpoint on its released tensors to verify our interpretation of that architecture.

II. Related Work

Witt and Young introduced phone-level pronunciation confidence for interactive language learning [1]; later DNN-GOP methods replaced generative likelihoods with neural posteriors [2]. SpeechOcean762 supplies a reproducible, expert-rated non-native English benchmark and a Kaldi recipe [8]. MFA packages a Kaldi-based text–speech alignment workflow for broader use [9], [10].

Cao et al. formulate scalar and vector CTC-GOP features in alignment-based and alignment-free forms [5]. Their strongest vector features can be consumed by GOPT, a compact Transformer that combines acoustic GOP features, canonical phone embeddings, learned position embeddings, and phone-, word-, and utterance-level supervision [7]. The original GOPT input is an

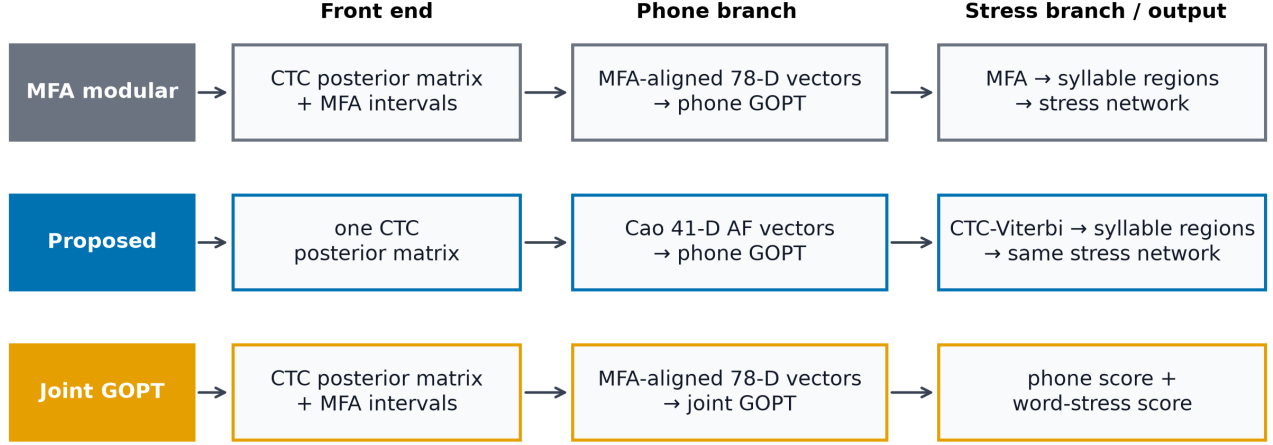


Fig. 1. The three controlled complete systems. The proposal uses Cao’s path-marginalized features for phone scoring and computes a single transcript-conditioned Viterbi path only for syllable stress.

84-dimensional Kaldi GOP vector. We retain its model topology but use controlled CTC-derived vectors so the three systems share one posterior source.

Lexical stress depends on relative syllabic cues rather than one absolute measurement [3]. Yarra et al. examine boundary sources and contextual features for automatic stress detection [4]. Mallela et al. model a word as a variable-length sequence of syllables with recurrent networks and a post-processing constraint that selects one primary stress [6]. Our local stress checkpoint follows those architectural ideas. It is an adaptation, not their unreleased ISLE checkpoint or an exact reproduction of their corpus experiment.

III. Three Assessment Systems

A. Shared CTC posterior matrix

All systems use the Wav2Vec 2.0 XLSR-53 large phone recognizer released by Cao et al., fine-tuned on LibriSpeech train-clean-100. It emits log posteriors $\ell_{t,k} = \log p(k | \mathbf{x}, t)$ for 39 ARPAbet phones plus CTC blank. This common encoding is charged once to every complete pipeline. The experimental branches differ after this matrix.

B. Proposed alignment-free phone branch

Let $\mathbf{y} = (y_1, \dots, y_U)$ be the canonical phone sequence and

$$G(\mathbf{y}) = -\log \sum_{\pi: B(\pi)=\mathbf{y}} \exp \sum_t \ell_{t, \pi_t} \quad (1)$$

be its CTC negative log likelihood, where B removes blanks and repeated labels. For each canonical position u , Cao’s vector concatenates $G(\mathbf{y})$ with 40 local contrasts

$$r_{u,a} = G(\mathbf{y}_{u \leftarrow a}) - G(\mathbf{y}), \quad (2)$$

covering deletion and all 39 phone substitutions. The result is a 41-D vector per phone and sums over all legal CTC paths; it contains no boundary estimate. We implement the official ordering and verify its loss values against exhaustive path enumeration on small lattices.

These vectors enter a phone-only GOPT: a linear feature projection, a 40-way canonical-phone projection, learned positions, five learned class tokens, three 24-dimensional one-head Transformer blocks, and a phone regression head. The class tokens preserve the official topology even though auxiliary losses are absent.

C. Embedded CTC-Viterbi stress branch

Stress features require regions. We expand the canonical sequence into

$$\mathbf{z} = (\phi, y_1, \phi, y_2, \dots, \phi, y_U, \phi) \quad (3)$$

and decode

$$V_t(s) = \ell_{t, z_s} + \max_{r \in \mathcal{P}(s)} V_{t-1}(r). \quad (4)$$

The predecessor set permits a self-loop, one-state advance, and a two-state skip into a phone only when adjacent labels differ. Repeated phones must pass through blank. Backtracking provides one contiguous region per canonical phone in $\mathcal{O}(TU)$ time. The implementation has explicit repeated-phone and small-lattice tests.

Maximal-onset syllabification groups the phone regions, and the vowel interval defines the nucleus. The local 42,061-parameter scorer consumes 38 features per syllable (19 acoustic and 19 contextual), uses masked LSTM layers of 64, 32, and 16 units, a four-dimensional attention layer, sigmoid scores, and a word-level argmax. The comparison uses only this neural output (neural_weight=1.0); the deployed acoustic-prominence ensemble is disabled.

D. MFA modular baseline

MFA 3.4.2 supplies word and phone intervals. For each phone interval we average the 39 non-blank CTC log posteriors and concatenate 39 target-relative log-posterior differences, yielding a 78-D aligned vector. A phone-only GOPT of the same topology as above predicts the phone grade. The same fixed 38-D stress feature code and sequential checkpoint use MFA rather than Viterbi regions. Thus MFA is charged for one alignment, one CTC encoding, a phone head, and the complete stress branch.

MFA phone sequences are globally matched to expert canonical sequences with Levenshtein dynamic programming. Every diagonal match is retained, including substitutions; an MFA interval is then labeled with the corresponding expert canonical phone. Insertions and deletions alone are excluded. This avoids artificially removing pronunciation errors merely because MFA chose a different phone symbol.

E. Joint GOPT comparator

The third system uses the same MFA-aligned 78-D vectors but retains GOPT’s multi-task heads: phone accuracy; word accuracy, stress, and total; and five utterance dimensions. It therefore produces both requested scores with one model and no separate stress feature network. This is a controlled adaptation of the published architecture, not the unchanged official Kaldi-front-end pipeline. We validate the latter separately in Sec. V-A.

IV. Experimental Design

A. Corpus, coverage, and training

SpeechOcean762 contains 5,000 English utterances from 250 Mandarin-speaking learners, split into speaker-disjoint train and test partitions of 2,500 utterances and 125 speakers each [8]. Five experts provide phone accuracy grades from 0 to 2. Word stress is 10 when accepted and 5 when incorrect. We use the official split.

Cao features are available for 47,076 training and 47,369 test phones. MFA exports 2,462 training and 2,447 test TextGrids with `-ignore_oovs`; diagonal mapping retains 46,064 and 46,119 phones, respectively. All three phone systems are evaluated on the same 46,119 test phones. The alignment-free model uses every available training phone, while the MFA-dependent systems use 46,064. This 2.2% coverage advantage is part of the complete system comparison and a possible contributor to accuracy.

Feature families are normalized with one scalar mean and standard deviation estimated only on their training examples. Models train for 100 epochs with Adam (10^{-3} , $\beta_1 = 0.95$, $\beta_2 = 0.999$, weight decay 5×10^{-7}), the released warm-up/multi-step schedule, and batch size 25. We report mean and sample standard deviation over five deterministic, independent initializations; predictions are not ensembled.

TABLE I

Phone accuracy on 46,119 common test phones. Learned-model values are mean $\pm$ sample standard deviation over five seeds.

Metric	MFA	Proposed	Joint GOPT
PCC	0.413 $\pm$.013	0.645 $\pm$.009	0.418 $\pm$.005
SRCC	0.381 $\pm$.004	0.432 $\pm$.003	0.378 $\pm$.003
MSE	0.112 $\pm$.002	0.079 $\pm$.002	0.111 $\pm$.001

B. Endpoints and inference

Phone endpoints are Pearson (PCC), Spearman (SRCC), and mean-squared error (MSE) against expert phone grades. Run-matched PCC differences summarize model initialization variation.

The stress set contains 2,354 common polysyllabic words from 1,578 utterances and all 125 test speakers: 2,282 accepted-stress tokens and 72 stress errors. Every system is evaluated on this same set. The primary common endpoint is area under the receiver operating characteristic curve (AUROC) for accepted versus incorrect stress. We additionally report PCC with the binary stress grade. For the two sequential branches, canonical-location accuracy is meaningful among words rated correct: when expert stress is accepted, a successful detector should select the dictionary-primary syllable. GOPT emits a scalar word-stress grade and does not produce a location.

For the two fixed sequential conditions, we resample speakers with replacement 10,000 times and recompute paired Viterbi-minus-MFA differences in AUROC and location accuracy. These are uncertainty intervals, not a preregistered non-inferiority test.

C. Equal-resource latency

Latency is measured on the same ARM64 macOS CPU, one request at a time. Every system uses four PyTorch intra-operation threads; MFA uses exactly one job. One CTC warm-up request is excluded. We record 100 complete front-end requests. Stress time is measured over the paired set and amortized over all 2,447 MFA-aligned test requests, assigning zero to requests without a scored polysyllable. A fresh one-job MFA corpus run takes 35.71 ms per requested input, including failures. No multi-worker alignment-only timing enters the complete comparison.

V. Results

Table I shows a stable advantage for Cao’s alignment-free features. The run-matched PCC gain is 0.232 ± 0.019 over modular MFA and 0.226 ± 0.009 over joint GOPT; every seed favors the proposal. This result also corrects the conceptual error of using one Viterbi path as the proposed phone score: Viterbi is absent from the phone branch.

The proposed fixed stress branch has 0.019 higher AUROC but 0.70 percentage points lower location accuracy than its MFA counterpart. The paired AUROC interval is $[-0.008, 0.053]$ and the location interval is

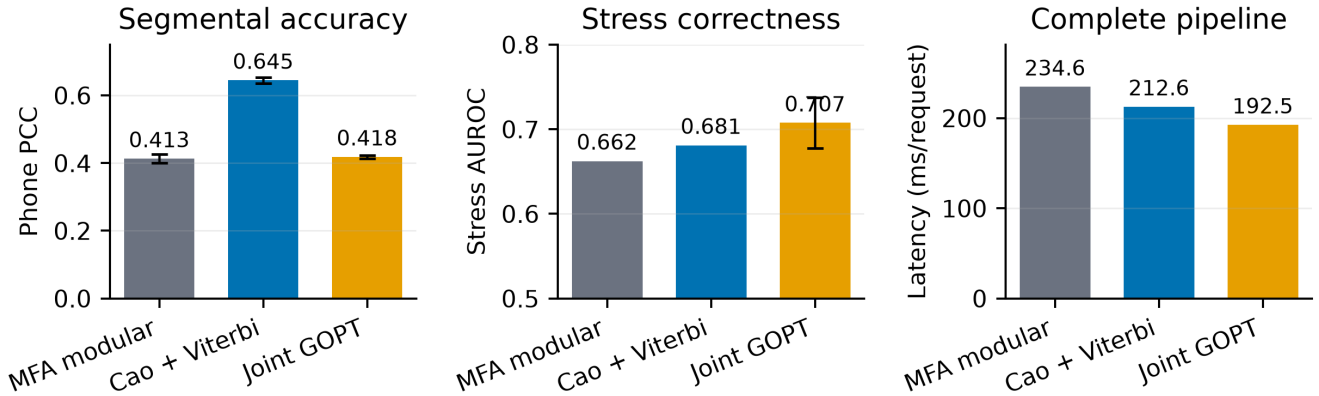


Fig. 2. Held-out phone PCC, stress-correctness AUROC, and complete single-request latency. Phone and joint-GOPT stress error bars are sample standard deviations across five seeds; the fixed stress bars have no initialization variance.

TABLE II
Word-stress accuracy on 2,354 common polysyllabic words.

Metric	MFA	Proposed	Joint GOPT
AUROC	0.662	0.681	0.707 ± .030
PCC	0.077	0.100	0.189 ± .017
Canonical location [†]	85.85%	85.14%	n/a

[†]Among 2,282 words rated stress-correct by experts.

TABLE III
Mean complete-pipeline CPU latency (ms/request).

Component	MFA	Proposed	Joint GOPT
Shared CTC encoding	154.85	154.85	154.85
Phone vector/head	1.91	8.95	1.94
CTC-Viterbi	—	2.98	—
Stress scorer	42.15	45.79	—
External MFA	35.71	—	35.71
Complete	234.63	212.57	192.50

$[-0.018, 0.005]$; neither excludes zero. The data therefore show no statistically resolved boundary-source difference, not proven equivalence. Joint GOPT has the strongest mean scalar stress metrics, though its AUROC varies more across initializations and it cannot localize the predicted stress.

The proposal reduces complete latency by 22.06 ms, a $1.10\times$ speedup over modular MFA (Table III). Cao’s 40 alternative CTC losses cost more than reading MFA-aligned vector regions, and Viterbi adds 2.98 ms; both remain cheaper than external MFA. Joint GOPT is fastest overall ($1.22\times$ the modular baseline) because its single head replaces the 42–46 ms TensorFlow stress branch. It therefore offers the strongest speed and scalar stress score here, while the proposal offers substantially higher phone correlation and explicit stress location.

A. Official GOPT checkpoint check

We ran the unmodified official model, checkpoint, normalization constants, and released LibriSpeech test tensors. It obtains phone PCC 0.6183 over 47,369 phones and word-stress PCC 0.3245 over 15,967 words, closely matching the repository values 0.616 and 0.326. This confirms the implementation path used to derive our controlled model. Its 0.425 ms GOPT-head latency excludes the Kaldi GOP/alignment front end and is not used in Table III.

VI. Discussion and Limitations

The results support a division of labor. Marginalizing every CTC path is highly effective for phone scoring; selecting one canonical path is sufficient to localize syllable features without a resolved stress penalty. This is more precise than proposing Viterbi as a universally superior alignment or phone score. The third system also matters: external alignment does not automatically make a complete system slower if a joint head removes a costly downstream network.

Several limitations bound the claim. First, the local stress checkpoint is Mallela-inspired rather than their ISLE model, and SpeechOcean supplies only a correct/incorrect word grade, not the realized destination of an incorrect stress. Second, only 72 common polysyllabic words carry a stress-error label; AUROC is consequently uncertain. Third, the alignment-free phone model sees 1,012 more training phones than MFA-dependent models. That coverage is a real system advantage but weakens a feature-only causal interpretation. Fourth, the controlled GOPT adaptation uses 78-D CTC vectors rather than its original 84-D Kaldi GOP front end, so its SpeechOcean row should not be compared directly with official LibriSpeech performance. Fifth, MFA is a baseline interval source, not manual boundary ground truth, and Viterbi timestamps should not be read as articulatory truth.

Finally, latency depends on hardware, implementation, batching, and parallelism. We intentionally compare one request, four common compute threads, and one MFA job. A production MFA pool or a batched CTC service would answer a different throughput question. Formal stress non-inferiority would also require a prespecified margin and a corpus with substantially more stress errors.

VII. Conclusion

We implemented the intended hybrid pipeline: Cao et al.’s alignment-free vector features score phones, while a transcript-conditioned CTC–Viterbi path exists only to provide regions for a Mallela-inspired stress network. Against a complete modular MFA baseline, this design improves phone PCC from 0.413 ± 0.013 to 0.645 ± 0.009 , shows no statistically resolved stress difference, and lowers single-request latency from 234.6 to 212.6 ms. A controlled joint-GOPT comparator is faster and stronger on scalar stress but substantially weaker on phone correlation and lacks stress location. The evidence therefore supports alignment only where the output requires it, with the claim restricted to the measured models, corpus, and execution protocol.

References

- [1] S. M. Witt and S. J. Young, “Phone-level pronunciation scoring and assessment for interactive language learning,” *Speech Communication*, vol. 30, no. 2-3, pp. 95–108, 2000.
- [2] W. Hu, Y. Qian, F. K. Soong, and Y. Wang, “Improved goodness of pronunciation (GOP) measure for phone-level pronunciation assessment using DNNs,” in *Proc. Interspeech 2015*, 2015, pp. 854–858.
- [3] D. B. Fry, “Experiments in the perception of stress,” *Language and Speech*, vol. 1, no. 2, pp. 126–152, 1958.
- [4] C. Yarra, M. K. Ramanathi, and P. K. Ghosh, “Comparison of automatic syllable stress detection quality with time-aligned boundaries and context dependencies,” in *Proc. 8th ISCA Workshop on Speech and Language Technology in Education (SLaTE 2019)*, 2019, pp. 79–83.
- [5] X. Cao, Z. Fan, T. Svendsen, and G. Salvi, “A framework for phoneme-level pronunciation assessment using CTC,” in *Proc. Interspeech 2024*, 2024, pp. 302–306.
- [6] J. Mallela, S. H. Aluru, and C. Yarra, “A comparative analysis of sequential models that integrate syllable dependency for automatic syllable stress detection,” in *Proc. Interspeech 2024*, 2024, pp. 3829–3833.
- [7] Y. Gong, Z. Chen, I.-H. Chu, P. Chang, and J. Glass, “Transformer-based multi-aspect multi-granularity non-native english speaker pronunciation assessment,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2022)*, 2022, pp. 7262–7266.
- [8] J. Zhang, Z. Zhang, Y. Wang, Z. Yan, Q. Song, Y. Huang, K. Li, D. Povey, and Y. Wang, “speechocean762: An open-source non-native english speech corpus for pronunciation assessment,” in *Proc. Interspeech 2021*, 2021, pp. 4099–4103.
- [9] M. McAuliffe, M. Socolof, S. Mihuc, M. Wagner, and M. Sonderegger, “Montreal forced aligner: Trainable text-speech alignment using Kaldi,” in *Proc. Interspeech 2017*, 2017, pp. 498–502.
- [10] D. Povey, A. Ghoshal, G. Boulianne, L. Burget, O. Glembek, N. Goel, M. Hannemann, P. Motlicek, Y. Qian, P. Schwarz, J. Silovsky, G. Stemmer, and K. Vesely, “The Kaldi speech recognition toolkit,” in *Proc. IEEE Workshop on Automatic Speech Recognition and Understanding (ASRU 2011)*, 2011.